

Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	Smart Lender – AI-Powered Loan Eligibility Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines the current limitations of the Smart Lender application, identifies areas for future scalability, and presents a roadmap for enhancing the system. Although the current implementation successfully demonstrates an AI-powered loan eligibility prediction platform running on a local development environment, the application has been designed with modularity and extensibility in mind, allowing it to evolve into a production-ready solution capable of serving financial institutions on a larger scale.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local Development Environment	The application currently runs on the Flask development server (<code>localhost</code>) and is intended for demonstration purposes only. It cannot efficiently handle multiple concurrent users or production-level traffic.	High
2	No Persistent Database	User information and prediction history are not stored permanently. Once the application is restarted, previous prediction records are unavailable.	High
3	Limited User Authentication	The current system does not include user registration, login, or role-based access control, allowing unrestricted access to the application.	High
4	Static Training Dataset	The machine learning model is trained on a fixed historical dataset and does not automatically learn from newly submitted loan applications.	Medium
5	Local File-Based Model Storage	The trained model and preprocessing artifacts are stored as local files (<code>.pkl</code>), making collaborative deployment and centralized model management difficult.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	Application Deployment	Runs locally using the Flask development server.	Deploy the application using Gunicorn or uWSGI behind an Nginx web server on a cloud platform such as AWS, Azure, or Render for improved scalability and availability.

2	Database Management	No persistent database is used.	Integrate PostgreSQL or MySQL to securely store user accounts, loan applications, prediction history, and system logs.
3	Batch Processing Performance	Batch prediction processes uploaded CSV files sequentially.	Implement asynchronous task processing using Celery with Redis to improve performance when handling large datasets.
4	Security	Basic input validation is implemented without authentication or API protection.	Introduce secure user authentication, role-based access control, encrypted credentials, environment variables, HTTPS, and API rate limiting.
5	Machine Learning Model Management	The model must be retrained manually whenever new training data becomes available.	Develop an automated model retraining pipeline capable of periodically updating the deployed model using newly collected loan application data.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Database Integration and User Authentication	Week 2	Integrate a relational database and implement secure user authentication to support persistent data storage, user management, and application history.
Phase 2	Cloud Deployment and Continuous Integration	Week 3	Containerize the application using Docker and deploy it to a cloud platform with automated testing and continuous integration pipelines for reliable deployment.
Phase 3	Third-Party Financial API Integration	Week 4	Integrate external credit bureau APIs and financial services to retrieve verified credit scores and financial information automatically, improving prediction reliability.
Phase 4	Advanced Explainable AI and Model Management	Week 5-6	Enhance prediction transparency using SHAP or LIME explanations and develop an administrative dashboard for automated model retraining, monitoring, and performance management.